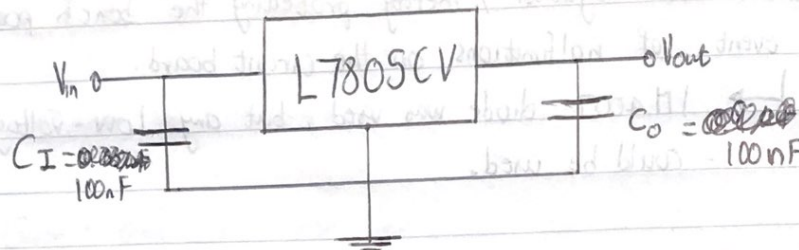


TEST 4 PREP 2025

DEMO 1 stuff

1. DESIGN OF VOLTAGE REGULATORS

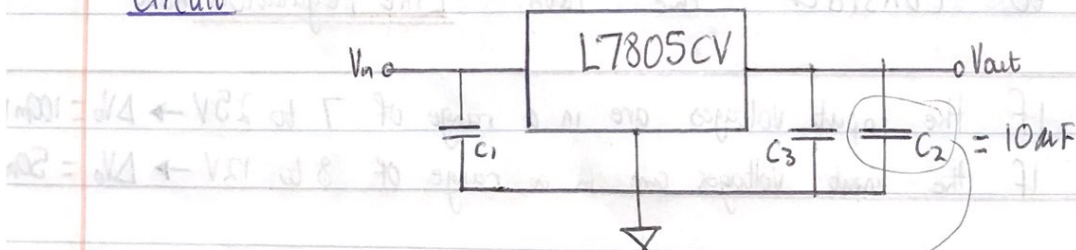
5V Regulator L7805CV



Input Capacitor (C_I) → Stabilizes the input voltage and filters out high frequency noise.

Output Capacitor (C_O) → Improves the transient response and output stability by reducing the output voltage ripple and noise caused by sudden load changes.

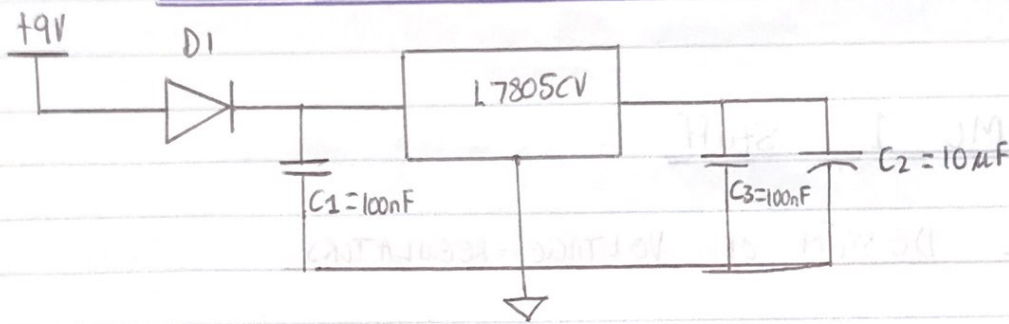
The role of the Additional Output Capacitor C_2 on the 5V regulator circuit



Absorbs voltage spikes resulting from a switching load quickly, thus protecting the circuit from unwanted transients.

→ It is worth noting that the capacitor C_2 could have been left out because the L7805CV has its own internal frequency compensation built into its design.

The Role of the Diode D1



* The role of Diode D1 is to ensure the direction of current flow is toward the Regulator, thereby protecting the bench power supply in the event of malfunctions on the circuit board.

→ 1N4007 diode was used, but any Low-voltage drop diode could be used.

CALCULATIONS ASSOCIATED WITH L7805CV REGULATOR

IMB all properties of the Regulator we used should be that in Table 11 pg 15

Q1 Unintended Change in the input voltage on Bench Power Supply Vs change in Output Voltage of the Regulator :-

"Given :- Temperature remains constant at 25°C"

have to consider the " ΔV_o " Line regulation how

* If the input voltages are in a range of 7 to 25V $\rightarrow \Delta V_o = 100\text{mV}$

* If the input voltages are in a range of 8 to 12V $\rightarrow \Delta V_o = 50\text{mV}$

Example :- You powered the device using a ^{9V} power supply in labs. When plugging something in, you accidentally changed the output to 10V while the device was powered. What is the most you'd expect the output voltage of L7805 to change?

9V $\rightarrow$ 10V $\Delta V_o = 100\text{mV}$

Q2

Sudden change in LOAD current Vs the change in the Output Voltage ΔV_o

"Given :- Temperature remains constant at 25°C "

HAVE TO CONSIDER the " ΔV_o " LOAD REGULATION ROW

* IF the Load currents ranges from 5mA to 1.5A $\rightarrow \Delta V_o = 100\text{mV}$

* IF the Load currents ranges from 250mA to 750mA $\rightarrow \Delta V_o = 50\text{mV}$

Q3

Change in temperature Vs Output Voltage of Regulator

"Given : that the load current is $I_o = 5\text{mA}$ "

HAVE TO GO TO $\Delta V_o / \Delta T$ column :-

$$\frac{\Delta V_o}{\Delta T} = -1.1 \text{ mV}/^\circ\text{C}$$

Example :- What is the new output voltage if the temperature were to increase to 72°C .

$$\frac{V_f - V_i}{T_f - T_i} = -1.1$$

$$V_f = -1.1 \frac{\text{mV}}{^\circ\text{C}} [T_f - T_i] + V_i$$

$$V_f = \frac{-1.1}{1000} [72 - 25] + 5$$

$$V_f = 5.0517\text{V}$$

The start current doesn't really specify the initial output voltage, but know that from our project the desired output is

5V

Q4

Calculating the maximum temperature at which the regulator operates at to decide whether heat sink is needed or not?

* For $T_O - 220$ package that L7805CV that we use is in, the Thermal resistance junction - ambient constant is 50°C (from table 2 pg 5)

$$\theta_{ja} = 50^\circ\text{C}/\text{W}$$

$$\text{Power} = V_{\text{across Regulator}} \times I_{\text{max consumed by Load}}$$

$$\text{Power} = [V_{\text{in}} - V_{\text{out}}] \times I_{\text{max Load}}$$

* Say for instance, 9V is supplied from bench power supply and only 5V comes out of the regulator and the load current is 110mA

$$\text{Power} = [9 - 5][110 \times 10^{-3}] = 0.44 \text{ W}$$

$$T_{j\text{max}} = \text{Power} \times \theta_j + T_{\text{amb}}$$

let's say you are given an ambient temperature of $T_{\text{amb}} = 25^\circ\text{C}$

$$\text{Then } T_{j\text{max}} = [0.44 \text{ W} \times 50^\circ\text{C/W}] + 25^\circ\text{C} = 47^\circ\text{C}$$

* From table I (pg 5), the operating junction temperature range is "0 to 150°C ", and since for our design we got 47°C which falls within the range, no heat sink needed.

Q5 What is the absolute maximum input voltage that should be input to the L7805CV.

* The maximum input voltage for the L7805CV is dependent on the desired output voltage or the voltage at which the input voltage is to be regulated down to.

$$* \text{ For } V_{\text{out}} = 5 \text{ to } 18 \text{ V} \Rightarrow \text{MAX INPUT V} = 35 \text{ V}$$

$$* \text{ For } V_{\text{out}} = 20, 24 \text{ V} \Rightarrow \text{MAX INPUT V} = 40 \text{ V}$$

Q6 CALCULATING MINIMUM AND MAXIMUM POWER DISSIPATED BY L7805 REGULATOR

$$P_{\text{Delivered by Regulator}} = V_{\text{DROP ACROSS REGULATOR}} \cdot I_{\text{Load}}$$

$$P_{\text{Delivered by Regulator}} = (V_{\text{IN}} - V_{\text{out Regulated}}) \cdot \frac{V_{\text{to Load}}}{R_{\text{Load}}}$$

Given the input voltage range : $[V_{\text{min}} ; V_{\text{max}}]$ and Load Resistance (R_{Load}), the min and max power dissipated by the device are calculated as follow

$$* P_{\text{min}} = [V_{\text{min}} - V_{\text{out Regulated}}] \cdot \frac{V_{\text{out Regulated}}}{R_{\text{Load}}}$$

$$* P_{\text{max}} = [V_{\text{max}} - V_{\text{out Regulated}}] \cdot \frac{V_{\text{out Regulated}}}{R_{\text{Load}}}$$

Q7 CALCULATING POWER DISSIPATED IN THE REGULATOR

$$P_{\text{Dissipated in the Regulator}} = (V_{\text{supplied}} - V_{\text{out Regulated}}) (I_{\text{Load}} + I_{\text{d}})$$

↗ Quiescent Current

↗ Power dissipated in the regulator is equal to voltage drop across it multiply by total current (Current delivered to Load + Current used internally in the Regulator (I_{d}))

* For L7805CV, $I_{\text{d}} = 8 \text{ mA}$ (Table 11 pg 15)

e.g $V_{\text{in}} = 17.5 \text{ V}$, $R_{\text{Load}} = 39 \text{ ohms}$ and $I_{\text{d}} = 8 \text{ mA}$

$$P_{\text{Dissipated in Regulator}} = (17.5 - 5) \left(\frac{5}{39} + \frac{8}{1000} \right) = 1.7026 \text{ W}$$

The use of Jumper J30 ??

→ Allows test station to supply the student board with a nominal 9V.

The use of Jumper J31 ??

→ Allows 9V supply to regulator.

The use of Fuse F1 ??

→ Allows supply of 5V to STM32 board.

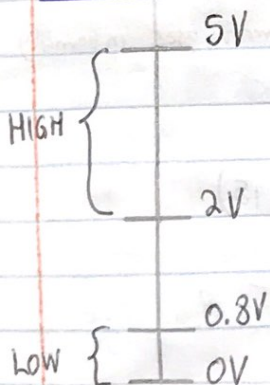
UART

"The UART shall operate at a baud rate of 57600 bps, with 8 data bits ; 1 parity bit, and one stop bit, using TTL signal levels. The parity used will be even parity"

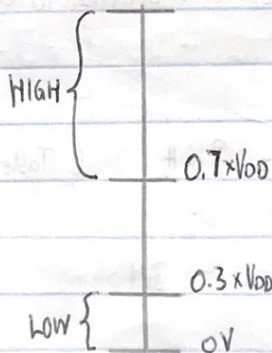
ACCEPTABLE TTL SIGNAL LEVELS DEPENDING ON WHETHER THE SIGNAL IS AN INPUT OR OUTPUT

Acceptable TTL Gate

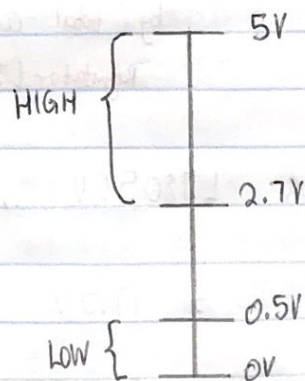
INPUT SIGNAL LEVELS



"NON-BOOT"



OUTPUT SIGNAL LEVELS

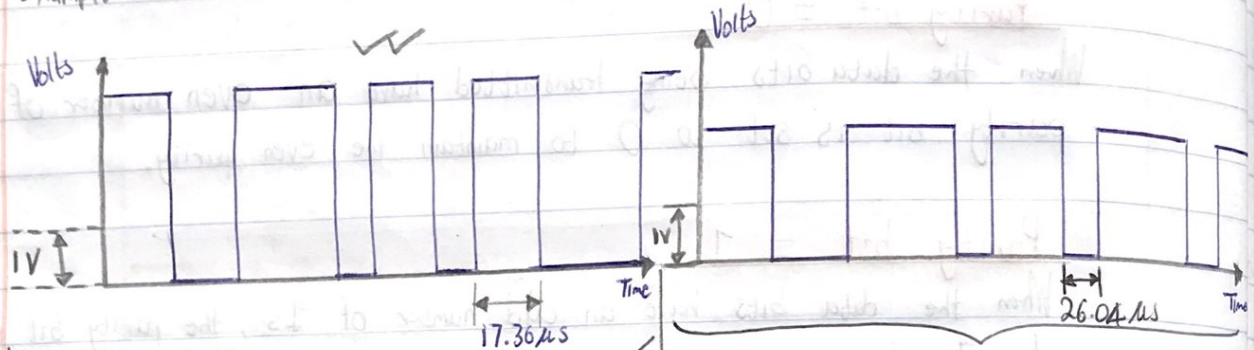


You gotta check if the High of the given signal is within the range of 2.7V to 5V or not ? ; Check if the Low is within the range of 0V to 0.5V

* FOR OUR SYSTEM THE HIGH SHOULD BE 3.3V and Low should be 0V.

* Baud Rate should be 57600 bps.

Example :-



$$\text{High Voltage} = (1 \times 3) + 0.3 = 3.3V \checkmark$$

$$\text{Baudrate} = \frac{1}{17.36 \times 10^{-6}} = 57603.7$$

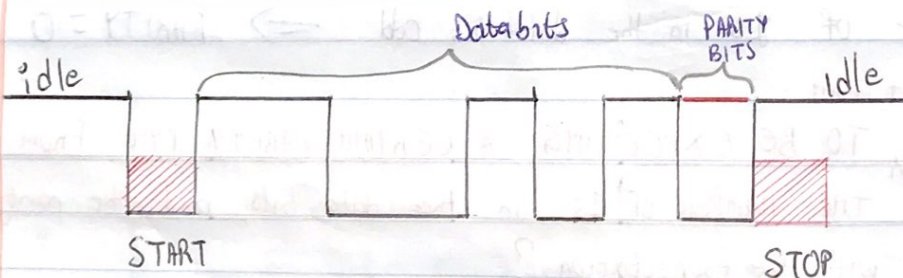
close enough to 57600 expected

$$\text{High Voltage} = (1 \times 2) + 0.3 \approx 2.3$$

↳ not 3.3V expected and not within the Required TTL High Voltage range.

$$\text{Baudrate} = \frac{1}{26.04 \times 10^{-6}} = 38402.5$$

UART FRAME FORMAT



Data bits → CAN BE 7 OR 8 depending on the specifications.

↳ NB! - On the UART FRAME FORMAT THE LEAST SIGNIFICANT BIT IS SENT FIRST, meaning when attempting to find the sent character you need to REARRANGE to MSB → LSB

PARITY BIT → An extra bit added to detect errors.

WHEN IS THE PARITY BIT EQUAL TO 0 and WHEN IS IT EQUAL TO 1??

Parity bit = 0

When the data bits being transmitted have an even number of 1s, the parity bit is set to 0 to maintain the even parity.

Parity bit = 1

When the data bits have an odd number of 1s, the parity bit is set to 1 to make the total number of 1s even

This is when the parity bit is said to be Even!

FOR ODD PARITY

* Number of ~~1s~~ 1s in the databits even $\Rightarrow$ PARITY = 1

* Number of 1s in the databits odd $\Rightarrow$ PARITY = 0

^{IT MEAN}
WHAT DOES IT MEAN TO BE EXPECTING A CERTAIN PARITY TYPE FROM A GIVEN SIGNAL BUT THE NUMBER OF 1s IN THE DATA BITS AND THE PARITY BIT DOESN'T AGREE WITH THE EXPECTATION?

e.g. Expecting Even parity

Number of 1s in databits = Even

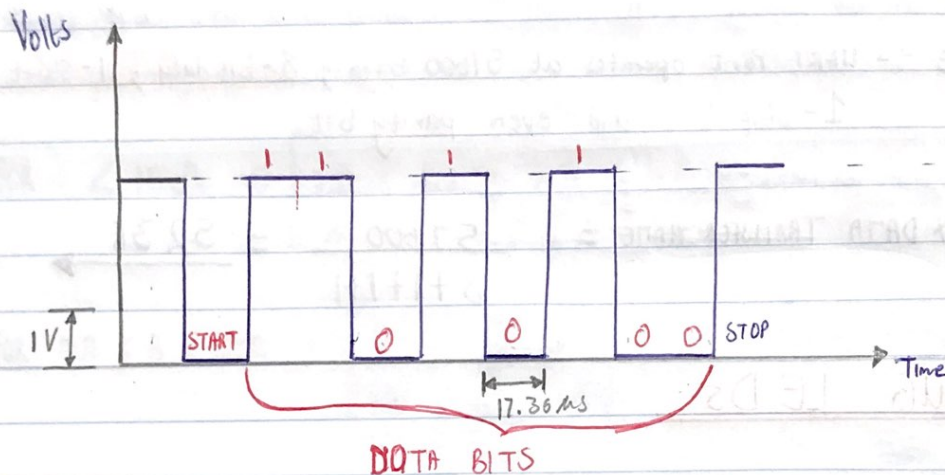
Parity bit = 1

→ This implies corruption of data during transmission.

WHAT DOES IT IMPLIES TO BE EXPECTING TO HAVE A CERTAIN PARITY BIT, BUT IS NOT AVAILABLE ON THE GIVEN ^{SIGNAL'S} UART FRAME FORMAT ??

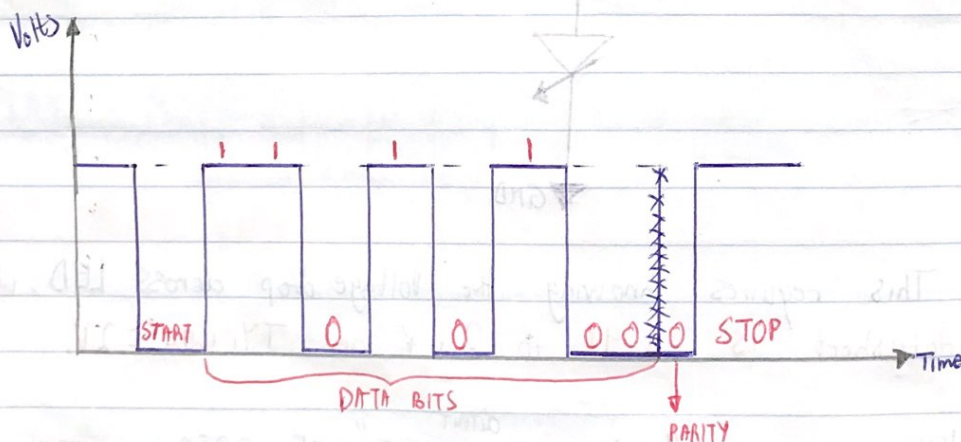
example: UART SPECS :- 57600 bps ; 8 data bits ; 1 parity bit and 1 Stop bit

Given signal :-



! (*) The above shows an invalid signal according to the expected specs, since the parity bit is expected but not there! and this means communication will fail, and not character or data will be transmitted

The CORRECT SIGNAL WOULD BE :-



Transmitted character : MSB → LSB
00101011 = +

CALCULATING MAXIMUM DATA TRANSFER RATE ON CHANNEL (Bytes/Second)

Given the UART Specs :-

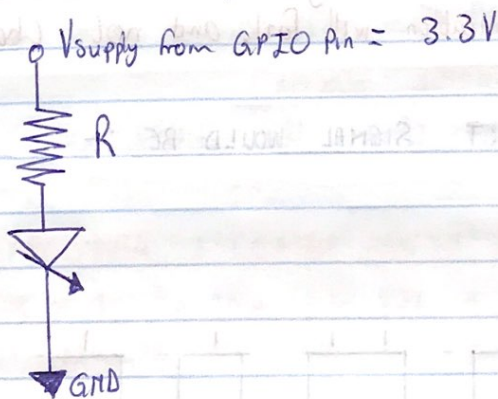
$$\text{MAX DATA TRANSFER RATE} = \frac{\text{BAUD RATE}}{\sum \text{of all bits}} = \frac{\text{BAUD RATE}}{\text{\# of data bits} + \text{START BIT} + \text{STOP BIT} + \text{PARITY BIT}}$$

Example :- UART Port operates at 57600 baud ; 8-bit data ; 1-start bit ; 1-stop bit and even parity bit

$$\text{MAX DATA TRANSFER RATE} = \frac{57600}{8+1+1+1} = 5236 \rightarrow$$

DC BUG LEDs

* The LED needs current limit through a series resistor, so as to not damage the processor output pins.



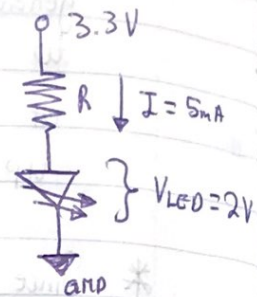
* This requires knowing the voltage drop across LED, which from the datasheet is 1.9V to 2.6V and TYPICAL = 2V.

* KNOWING THAT "OPERATIONAL CURRENT" OF GPIO pin goes up to 20mA.
 → NOT ABSOLUTE MAX! remember we never design for absolute MAX for anything.

EXAMPLE : CHOOSE Current to be 5mA and calculate R:-

$$\frac{V_{\text{supply}} - V_{\text{LED}}}{I} = R$$

$$\frac{3.3 - 2}{5 \times 10^{-3}} = 260 \Omega$$



STANDARD E12

FOR $R < 10 \Omega \Rightarrow [1.0 ; 1.2 ; 1.5 ; 1.8 ; 2.2 ; 2.7 ; 3.3 ; 3.9 ; 4.7 ; 5.6 ; 6.8 ; 8.2]$

FOR $10 \Omega \leq R < 100 \Omega \Rightarrow [10 ; 12 ; 15 ; 18 ; 22 ; 27 ; 33 ; 39 ; 47 ; 56 ; 68 ; 82]$

FOR $R \geq 1000 \Omega \Rightarrow [1000 ; 1200 ; 1500 ; 1800 ; 2200 ; 2700 ; 3300 ; 3900 ; 4700 ; 5600 ; 6800 ; 8200]$

$\Rightarrow$ CHOOSE THE CLOSEST !!

DEMO 2 STUFF

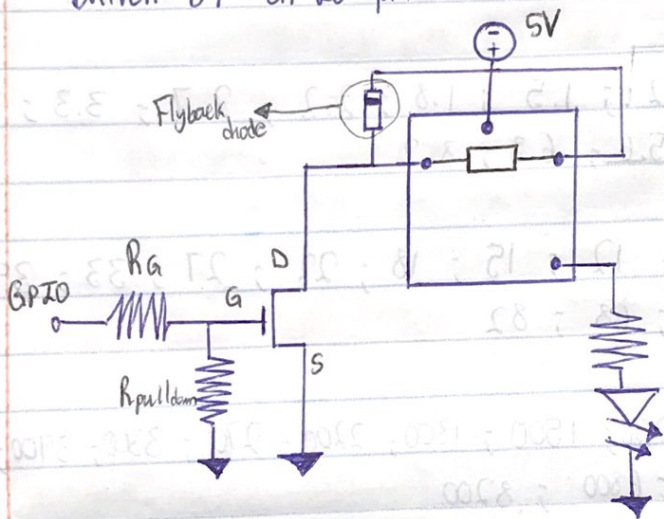
RELAY - MOSFET CIRCUIT

WHY RELAY CANNOT BE DRIVEN DIRECTLY FROM GPIO PINS?

1. The J107 series relay requires a minimum voltage (pick up voltage) of 3.75V in order for it to transition from one state to the other (ON/OFF), which is greater than 3.3V maximum voltage that can be supplied by GPIO pin.

2. The coil requires a current of 72.5 mA for it to be able to generate a magnetic field that will induce the movement of the switch to turn the load ON/OFF, But the GPIO pin can supply a maximum of 20 mA, and out of this 20 mA already some current is being used up by LED D2.

* Since the Relay Requires higher voltage and current than what the GPIO pin can supply ; A MOSFET IS USED SUCH THAT ~~IT IS~~ IT IS DRIVEN BY GPIO pin and connected to Relay to complete its circuit.

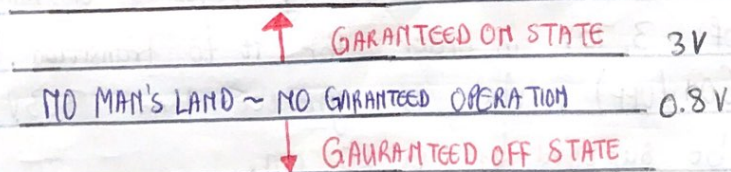


2N7000 MOSFET

* The mosfet is a voltage driven device which requires a very small current at the gate for it to turn on.

↳ Since MOSFET is a voltage driven device, the circuit design has to be decided based on the **GATE THRESHOLD VOLTAGE** :-

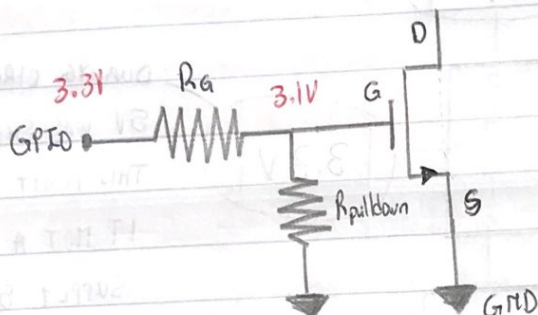
$$V_{gs(th)} = 0.8V \text{ to } 3V$$



PURPOSE OF SMALL SERIES RESISTOR R_G ON THE MOSFET CIRCUIT AND ITS VALUE CALCULATION

$R_G \rightarrow$ prevents in-rush current to the mosfet.

CALCULATING R_G :-



Since there's no specific requirement for gate current despite the fact that ~~it should be very small~~ IT SHOULD BE VERY SMALL

* CHOOSE $I = 1\text{mA}$

and turn on voltage of $V_{on} = 3.1\text{V}$

$$R_G = \frac{(3.3 - 3.1)\text{V}}{1\text{mA}} = 200\Omega$$

PURPOSE OF THE LARGE PULL-DOWN RESISTOR AND CALCULATING ITS VALUE

The large pull-down resistor ensures that the mosfet gate is firmly pulled to 0V (ground) when the control signal is inactive (low), and thus helps prevent accidental turn-on due to leakage currents or floating gate conditions.

CALCULATING $R_{pull-down}$:-

$R_{pull-down}$ is typically calculated based on the series gate resistor (R_G).

The rule of thumb is :-

$$R_{pull-down} \geq 10 \times R_G$$

To ensure a large enough value is used, chose to make it 10 times the minimum

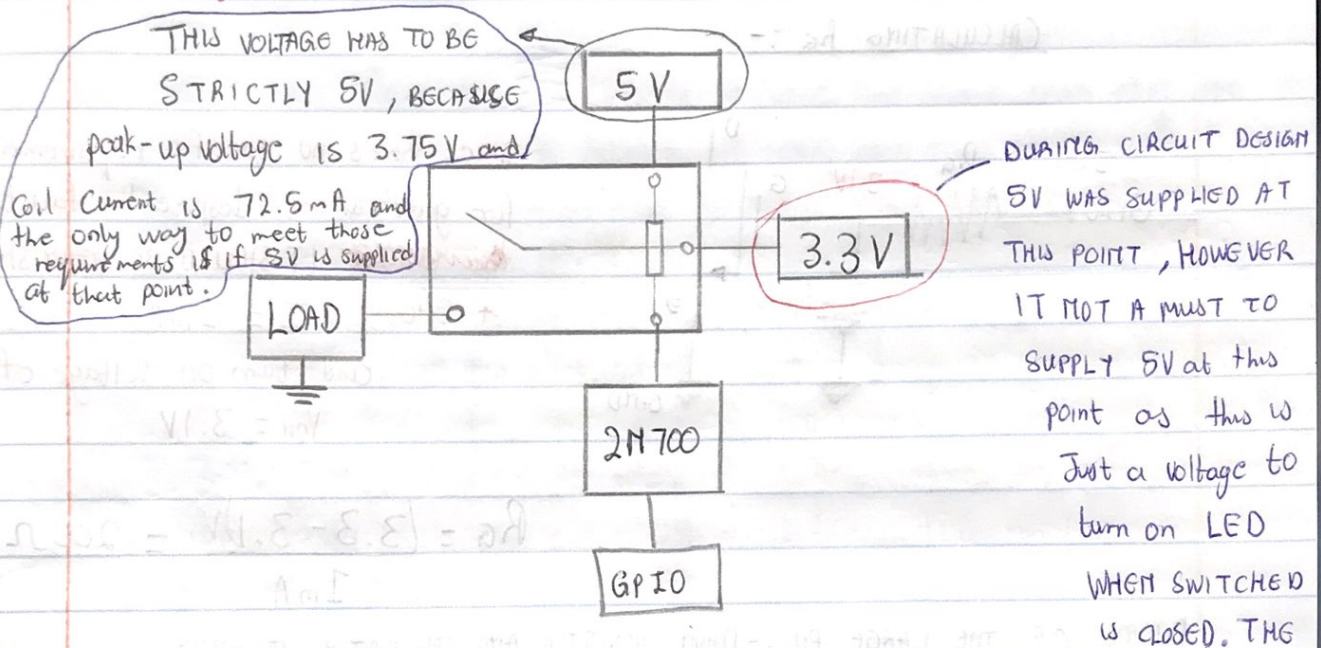
$$R_{pull-down} = 10 \times 10 \times R_G$$

$$R_{pull-down} = 10 \times 10 \times 200$$

$$R_{pull-down} = 2000\Omega$$

PURPOSE OF FLYBACK DIODE

⇒ The flyback diode prevents the high voltage spikes that gets generated when the relay switches ON; it also creates path and direction to which the energy that the coil stores can be dissipated to avoid damaging mosfet.

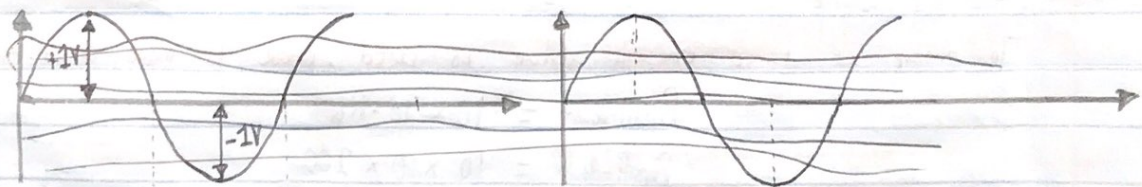


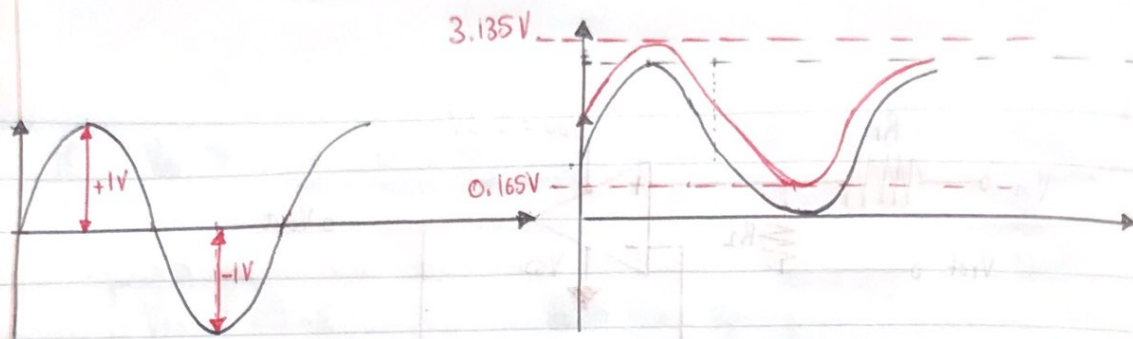
LED has Voltage drop of 1.9V to 2.6V and therefore 3.3V is more than enough.

ADC MEASUREMENTS

WHY IS THE OP-AMP CIRCUIT NEEDED?

* The given signal is Alternating signal and includes negative values. However, ADC cannot measure negative voltages and therefore need to scale the values to be in the range of 0-3.3V. Also need to make sure that the full ADC resolution is used to ensure accuracy of the results. Hence, adding offset and multiplying by the gain ~~to~~ through OP-AMP.





Even though the ADC input voltage range is 0-3.3V but design for absolute maximum values is avoided. This is ~~because~~ to avoid "Clipping" ~~and~~ due to component tolerances introducing variations that pushes the values beyond the expected range hence damaging the GPIO pins.

Design Range is $[0.165V ; 3.135V]$ instead of $[0V ; 3.3V]$

CALCULATING GAIN AND OFFSET

Lower peak Amplitude : Desired Lower Peak Amplitude

$$-1V \Rightarrow 0.165V$$

Higher Peak Amplitude : Desired Higher Peak Amplitude

$$1V \Rightarrow 3.135V$$

$$\begin{matrix} V_{in} & V_{out} \\ [-1V ; 0.165V] \end{matrix}$$

$$[1V ; 3.135V]$$

$$V_{out} = \text{Gain} \cdot V_{in} + V_{offset}$$

$$V_{out} = m \cdot V_{in} + b$$

$$0.165 = m(-1) + b$$

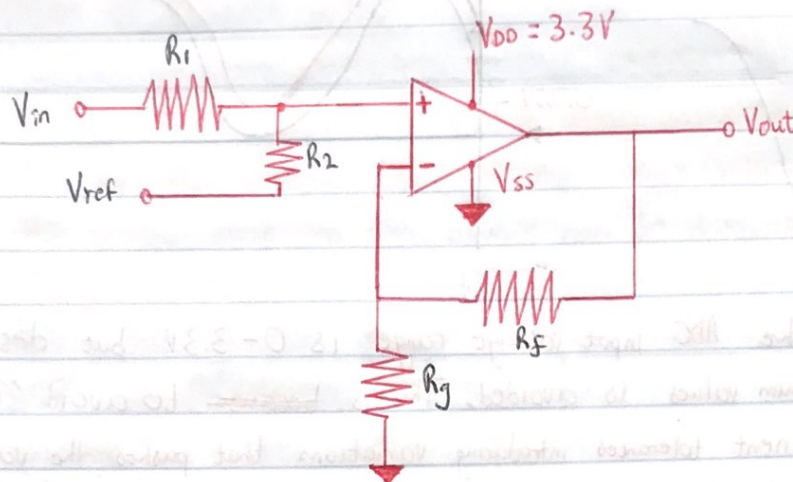
$$b = 0.165 + m \quad \text{--- (1)}$$

$$3.135V = m(1) + b \quad \text{--- (2)}$$

$$3.135 = m + 0.165 + m$$

$$m = 1.485$$

$$b = 1.65$$



WHAT SHOULD THE SUPPLY VOLTAGE V_{DD} BE ? and WHY?

The datasheet specifies the range of input supply voltage to be 2.7V to 6.0V, HOWEVER in our case V_{DD} needs to be strictly set to 3.3V. This is to ensure that should there be connection errors leading to saturation of the output, the ADC maximum input voltage is not exceeded! This is clearly explained by the equations used to find the output range at different conditions depending on the value of V_{DD} .

~~Maximum output voltage swing~~ ~~From DC characteristics Table - Output Section~~
 → From DC characteristics Table - Output Section

MAXIMUM OUTPUT VOLTAGE SWING

CONDITIONS

- * $V_{OL} = V_{SS} + 15\text{mV}$; $V_{OH} = V_{DD} - 20\text{mV}$ @ $R_L = 25\text{k}\Omega$ to V_L ; output drive = 0.5V
- * $V_{OL} = V_{SS} + 45\text{mV}$; $V_{OH} = V_{DD} - 60\text{mV}$ @ $R_L = 5\text{k}\Omega$ to V_L ; output drive = 0.5V

LINEAR OUTPUT VOLTAGE SWING

- * $V_{OUT,LOW} = V_{SS} + 100\text{mV}$; $V_{OUT(HIGH)} = V_{DD} - 100\text{mV}$ @ $R_L = 25\text{k}\Omega$ to V_L ; $A_{OL} \geq 100\text{dB}$
- * $V_{OUT,LOW} = V_{SS} + 100\text{mV}$; $V_{OUT(HIGH)} = V_{DD} - 100\text{mV}$ @ $R_L = 5\text{k}\Omega$ to V_L ; $A_{OL} \geq 95\text{dB}$

EXAMPLE :- Say for instance Question says it is MAXIMUM OUTPUT VOLTAGE SWING and the conditions are $R_L = 5\text{k}\Omega$ to V_L ; Output drive = 0.5V :-

If $V_{DD} > 3.3V$ let's say $5V$, then the voltage range going to ADC is :-

$$\left[\underset{0 + \frac{0.5}{1000}}{V_{SS} + 45mV} ; \underset{5 - \frac{60}{1000}}{V_{OH} - 60mV} \right]$$

$$\left[0.045V ; 4.94V \right]$$

$4.94V$ is way to greater than $3.3V$ (Rather $3.6V$ as specified by MCU datasheet) which is the MAXIMUM VOLTAGE THAT ADC peripheral of the microcontroller can handle. This then means you are running the risk of damaging the microcontroller's ADC peripheral.

WHAT SHOULD V_{ref} BE ? and WHY ??

⇒ V_{ref} was also set to $3.3V$, and this is because V_{ref} acts as a virtual ground and represents the offset voltage at which the sine wave should be symmetric to allow ^{usable} dynamic range. This V_{ref} must be stable and noise-free, as any noise or fluctuation would cause the entire signal to shift unpredictably, leading to inaccurate ADC readings or even clipping.

⊛ Therefore, $3.3V$ from our circuit board has already been well regulated, stabilized and incorporated with decoupling capacitors, hence its use for V_{ref} .

Finding VALUES FOR R_1 ; R_2 ; R_f and R_g ??

* R_1 and R_f should be given, e.g assume $R_1 = R_f = 10k\Omega$

* $R_2 = \frac{V_{ref} \times R_1 \times m}{b}$ where $m = \text{Gain}$; $b = V_{offset}$

* $R_g = \frac{R_2 \times R_f}{m \times (R_1 + R_2) - R_2}$

ADC, Setup and Sampling Data Management

TIM TRIGGERED ADC IN DMA MODE WAS USED → Allows time regulated ^{CONTINUOUS} sampling and storing of data to memory without CPU intervention.

DMA Advantages

- Allows ADC to directly transfer the data to memory without CPU intervention. [Peripheral to memory]
- Allows continuous high-speed ADC conversions without CPU intervention, ideal for scanning multiple ADC channels since it can automatically transfer data from multiple channels to memory.

WHAT SHOULD SAMPLING TIME BE SET TO? and WHY?

⇒ Needs to be set to highest available sampling time, to allow transient effects on the signal to settle.

CALCULATING SAMPLING FREQUENCY AND NUMBER OF SAMPLES??

Nyquist theorem:- $f_{\text{sampling}} \geq 2 \times f_{\text{signal}}$

e.g Say $f_{\text{signal}} = 50 \text{ Hz}$
 $f_{\text{sampling}} = 200 \times 2 \times 50 \text{ Hz} = 20 \text{ KHz}$

$$\text{Number of Samples} = \frac{f_{\text{sampling}}}{f_{\text{signal}}}$$

$$\text{Number of samples} = \frac{20000}{50} = 400 \text{ samples}$$

Buffer Size $\geq 2 \times \text{Number of samples}$ → ensures enough cycles of signal's wave are fed to ensure stability.

CALCULATING COUNTER PERIOD (ARR) AKA TIMER PERIOD

$$ARR + 1 = \frac{\text{Internal Clock Frequency}}{(\text{Prescaler} + 1) \cdot f_{\text{sampling}}}$$

CALCULATING CONVERSION TIME (TIME IT TAKES TO COMPLETE ONE FULL SEQUENCE OF BOTH SIGNALS)

$$T_{\text{conv}} = \frac{\text{Sampling Time} + 12.5}{\text{ADC CLOCK}}$$

$$\therefore \text{Total } T_{\text{conv}} = 2 \times T_{\text{conv}}$$

ADC VALUE $\leftrightarrow$ VOLTAGE

$$\text{ADC VALUE} = \frac{V_{\text{in}}}{V_{\text{ref}}} \times \text{ADC Resolution}$$

$$\text{ADC VALUE} = \frac{V_{\text{in}}}{V_{\text{ref}}} \times (2^n - 1)$$

CALCULATING PEAK-TO-PEAK VALUES

WHY NOT USING $2 \times V_{\text{max}} = V_{\text{pp}}$, even though it is correct mathematically/theoretically??

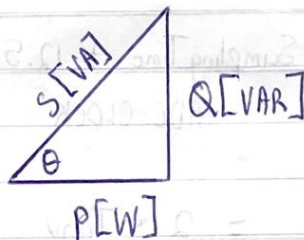
$\Rightarrow V_{\text{pp}} = 2 \times V_{\text{max}}$ assumes that the wave is perfectly centered at 0V ~~zero~~ or whatever offset (i.e. no asymmetry). But in reality the waveform may not ^{synchronous} (due to gain error; offset; load, or temperature drift) therefore even small offsets cause large errors when computing peak-to-peak without subtracting min from max.

THEREFORE CORRECT METHOD :- $V_{\text{pp}} = V_{\text{max}} - V_{\text{min}}$

DEMO 3 STUFF

WHY MEASURE THE PHASE DIFFERENCE?

→ Voltage and current in an electrical system don't peak at exactly the same time whenever there's any reactance in the load (even though their input parameters are same i.e. frequency and V_{pp}). Therefore, knowing phase angle or phase difference allows us to differentiate on how much power is actually delivered "Real power" and how much is just "Reactive Power".



Apparent Power : $S = V_{rms} \cdot I_{rms}$

Real Power : $P = S \cos(\theta)$

Reactive Power : $Q = S \sin(\theta)$

WHY DO WE NEED POWER FACTOR? [AKA Efficiency : FRACTION OF USEFUL POWER] ~ Percentage or Ratio of Real Power!!

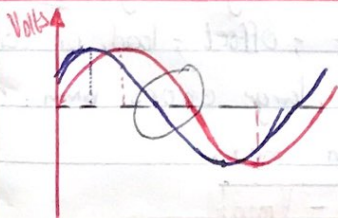
→ Power factor tells us the fraction of the Apparent Power (S) that is actually doing useful work :-

$$PF = \frac{P_{real}}{P_{apparent}} = \frac{S \cos \theta}{S} = \cos \theta$$

$PF = 1 \rightarrow$ All supplied energy is doing useful work.

$PF < 1 \rightarrow$ Some energy is lost to sustaining magnetic / electric fields.

CALCULATING PHASE ANGLE

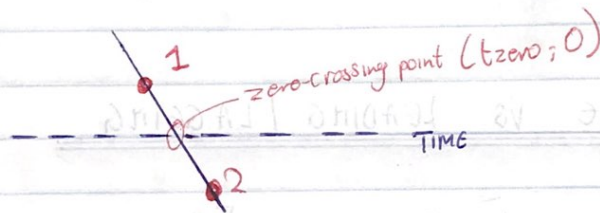


$$PF = \frac{P_{real}}{P_{apparent}} = \frac{P_{real}}{\sqrt{(P_{real})^2 + (P_{reactive})^2}} = \frac{S \cos \theta}{\sqrt{(S \cos \theta)^2 + (S \sin \theta)^2}} = \cos \theta$$

* To get phase angle, it is necessary to find the time difference between two points of the given waves. WHY ZERO CROSSING POINTS HIGHLY RECOMMENDED OVER THE PEAKS?

⇒ At sine wave's peaks the slope is zero, any small voltage noise or ADC quantization error will translate into a large uncertainty in the inferred peak time, thus inaccurate calculations, whereas at zero crossings waveform's slope is maximal, so the voltage uncertainty produces much smaller timing error and this is ideal for simple linear interpolation.

FINDING ZERO CROSSING TIME :-



point 1 : $(t_1 ; y_1)$

zero-crossing point : $(t_{zero} ; 0)$

point 2 : $(t_2 ; y_2)$

interpolation :
$$\frac{t_{zero} - t_1}{t_2 - t_1} = \frac{0 - y_1}{y_2 - y_1}$$

$$t_{zero} = \frac{-y_1}{y_2 - y_1} (t_2 - t_1) + t_1$$

$$t_{zero} = t_1 - \frac{y_1}{y_2 - y_1} (t_2 - t_1)$$

CHANGE IN TIME (Δt) ??

$$\Delta t = t_{zero, voltage} - t_{zero, current}$$

$\frac{P_{real}}{P_{real} + P_{reactive}}$



NOT

CALCULATING PHASE ANGLE ??

$$\phi = \omega \cdot \Delta t$$

$$\phi = 2\pi f_{\text{signal}} \cdot \Delta t$$

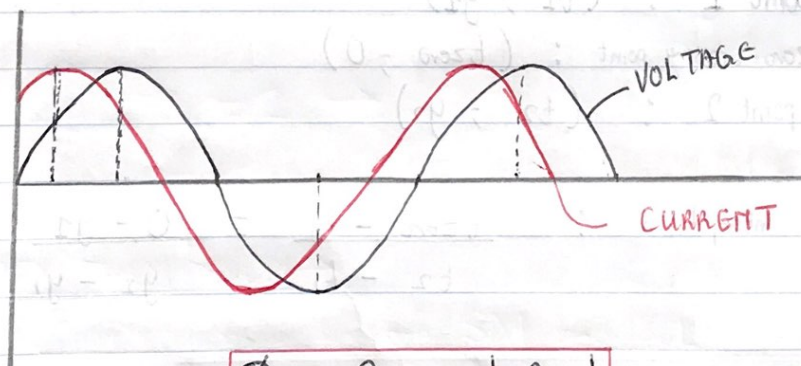
in Normal way :- $\phi = \theta_v - \theta_i$

WHY PHASE ANGLE HAD TO BE RESTRICTED TO $\pm 90^\circ$??

$\Rightarrow$ Phase angle has to be restricted to $\pm 90^\circ$ to ensure that the real power stays positive all the time. Otherwise, if it becomes negative it will seem as if the load is capable of generating the power whereas it only consumes it.

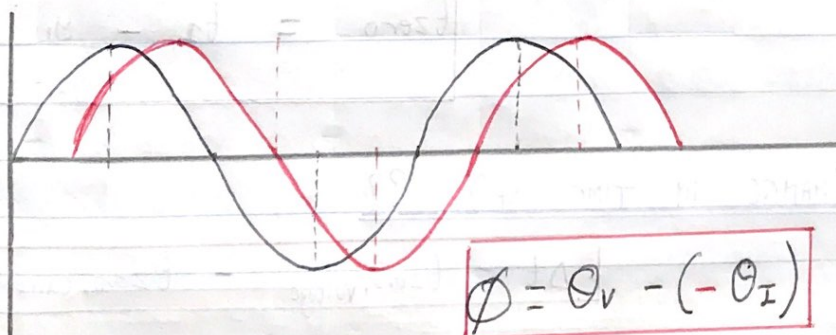
PHASE ANGLE VS LEADING / LAGGING

CASE 1: CURRENT LEADING VOLTAGE [PHASE ANGLE < 0]



$$\phi = \theta_v - |\theta_i|$$

CASE 2: VOLTAGE LEADS CURRENT [PHASE ANGLE > 0]



$$\phi = \theta_v - (-\theta_i)$$

OLED

* Uses I2C interface :-

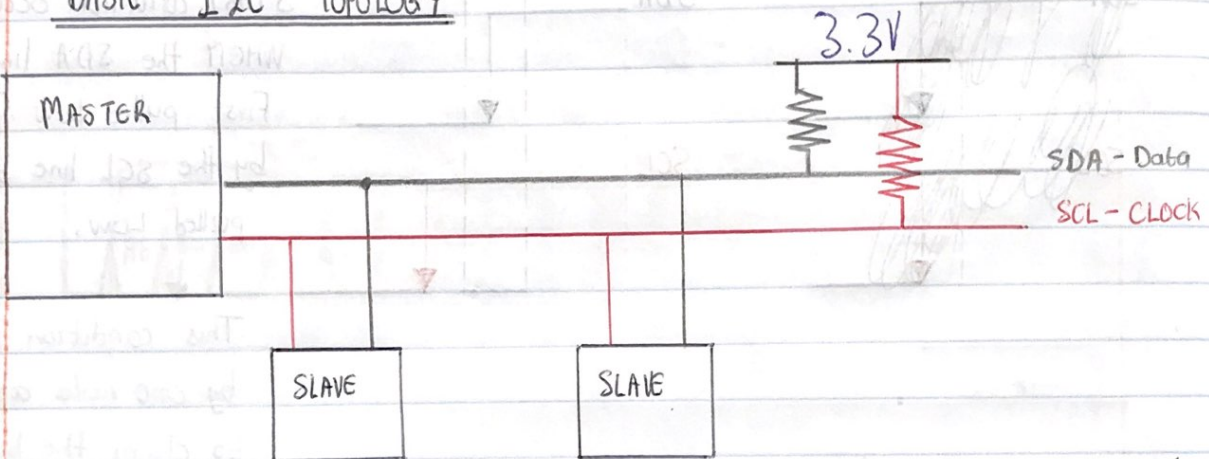
SDA → Serial Data

↳ Carries the data bits, one at a time, bi-directionally.

SCL → Serial Clock

↳ Carries the clock pulses, always generated by the master.

BASIC I2C TOPOLOGY



* In NORMAL "IDLE" state of the I2C (i.e. when ~~no~~ communication occurs)
⇒ SDA and SCL lines are pulled High, by the pull-up resistors shown on the topology above.

WHY ARE PULL-UP RESISTORS REQUIRED FOR CLOCK AND DATA LINES OF I2C PORT

⇒ The devices (slaves) connected to I2C port pull the SDA and SCL lines low, but they cannot drive it high. Therefore, the pull-up resistors ensure that the bus lines always return to the logic high level when idle (no communication).
↳ Without pull-up resistors the SDA and SCL lines would just float, leading to undefined logic levels and communications.

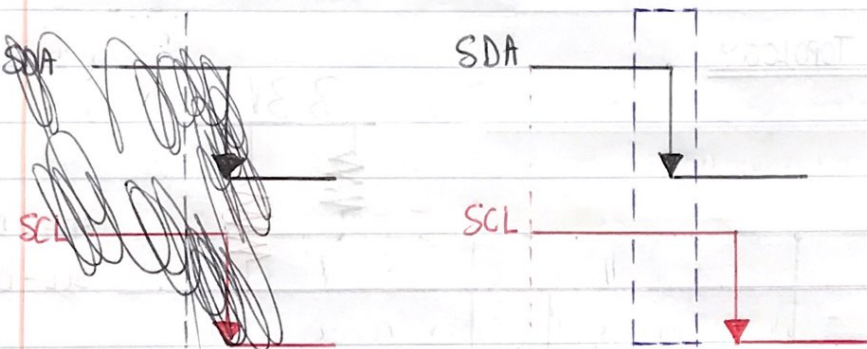
* The range of these pull-up resistors : $[1K\Omega \text{ to } 10K\Omega]$

* Higher pull-up resistance → LOWER BUS SPEED, and lower resistance → FAST SPEED

I2C MODES OR SPEEDS

I2C MODE	SPEED
Standard Mode	100 KHz
Fast Mode	400 KHz
Fast Mode plus	1000 KHz

START CONDITION FOR I2C

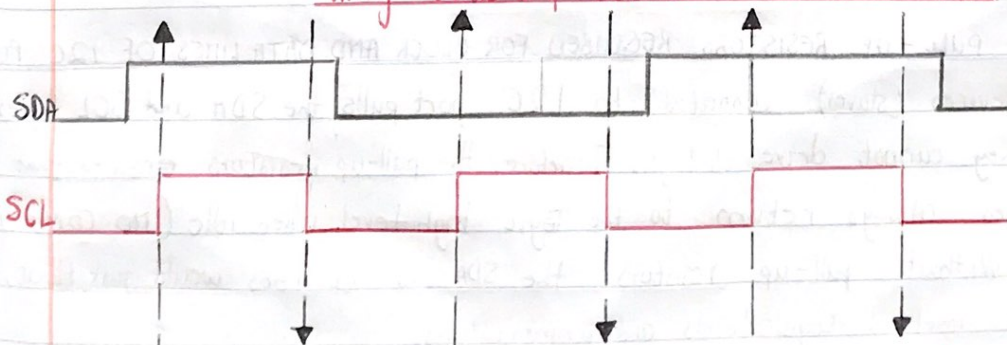


START CONDITION OCCURS WHEN the SDA line is first pulled low followed by the SCL line being pulled low.

↓
This condition is used by one node at a time to claim the bus (i.e. one slave at a time to prevent risk of contention)

DURING DATA TRANSMISSION

Timing relationship between SDA and SCL



▶ SDA does not change between SCL rising edge and SCL falling edge.

▶ During data transmission, SDA only transitions while SCL is LOW, IF it transitions while SCL is HIGH that is either a start or stop condition.